AATHITHYAN K
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SUMMARY

I'm an ECE student driven by a curiosity for how hardware and software come together to create real impact. My work spans embedded systems, microcontrollers, and sensor interfacing, supported by a strong base in Java development. I enjoy transforming ideas into functional systems that bridge electronics with intelligent software. I'm focused on building efficient, real-world solutions while continuously expanding my technical depth.

SKILLS

Technical Skills:
- HTML,CSS,PYTHON,JAVA,MYSQL,EMBEDDED SYSTEM

Tools:
- VSCODE,ANTIGRAVITY,GITHUB

Languages Known:
- TAMIL
- ENGLISH

EDUCATION

- V.S.B College of Engineering Technical Campus *(2023-2027)*
  B.E Electronics and Communication Engineering ***(GPA-8.4)*** ***Coimbatore,TamilNadu***
- Dr. V. Genguswamy Naidu Matric Hr. Sec. School *(2023)*
  *HSC -**(86%)*** ***Coimbatore,TamilNadu***

WORK EXPERIENCE

JAVA INTERN | QAROO INDIA PRIVATE LIMITED(*Coimbatore)* *JAN 2026*

- Developed and tested small-scale core Java applications, strengthening foundational OOP skills.
- Utilized debugging tools to isolate and resolve software bugs, improving program efficiency.
- Gained hands-on experience with modern software development workflows and industry best practices.

EMBEDDED SYSTEM INTERN |MANFREE TECHNOLOGIES(*Coimbatore)* *JUN 2025*

- **Embedded Firmware Development:** Programmed Basic **PIC microcontrollers** using **MPLAB X IDE** to interface with digital and analog sensors, building functional real-time embedded applications.
- **Hardware Simulation & Verification:** Modeled and tested fundamental circuit designs in **Proteus** prior to physical assembly, minimizing hardware design flaws and accelerating prototype development.
- **Circuit Diagnostics & Debugging:** Utilized oscilloscopes, multimeters, and software debugging tools to acquire signals, verify data protocols, and isolate firmware errors.

PROJECTS

Dynamic Weight Measuring and Data Logging System (**MOHLER MACHINE WORKS)**

- Developed an automated checkweigher conveyor belt system utilizing an Arduino, a 24-bit HX711 ADC, and a strain-gauge load cell to capture precise, real-time, in-motion weight profiling.
- Integrated two motors along with mechanical bearings and a decoupled battery power supply to isolate structural vibrations and electrical noise from the sensitive analog circuitry.
- Programmed synchronous firmware in C++ to smooth data streams via a digital moving average filter and automate localized data logging into structured CSV formats for downstream analysis.
- **Tools Used:** ARDUINO UNO and IDE, HX711 Module

CERTIFICATIONS

- Programming in JAVA | **NPTEL** *July - Nov 2025*
- Introduction to IOT | **NPTEL** *Jan -Apr 2025*
- Career Edge - Young Professional Course |**TCS ION** *Dec2024-Jan 2025*
- Basics Of PYTHON | **INFOSYS**
- Data Science for Beginners | **BOARD INFINITY**

ACHIEVEMENTS

- Participant in **SIMA TEXFAIR 2026** *Mar 2026*
  SIMA - The Southern India Mills Association
  Codissia Trade Fair Complex,Coimbatore
- Participant at **NVIDIA RTX AI PX DAY** *Jan 2026*
  NVIDIA – Asia Partner | *Codissia Trade Fair Complex, Coimbatore*
- Attended WorkShop on **IoT Automation using Raspberry PI and Node-Red** *Nov 2025*
  IIT Madras Research Park Chennai
- Published **PATENT** on **Surveillance of robotic boat using IoT and image processing** *Sept 2025*
  (IP Rights India **)**
- **International Conferenc**e on Sustainable Developments in STEM Apr 2025
  Online Mode | R&D Cell with *Gambella University, Ethiopia and Jai Shriram Engineering College Tirupur*
- Participated **PAPER PRESENTATION** AT **TECHAURA'25** *Apr 2025*
  KPR Institute of Engineering and Technology, Coimbatore